

Social constraints (Lecture) ~ Avel Guénin-Carlut ~ ActInf Social Sciences

Courses/ActiveInferenceForTheSocialSciences | Avel Guénin-Carlut ~ ActInf Social Sciences 2023 | 1:41:27

Hello and welcome everyone. It's October 18th, 2023. We're in the course Active Inference for the Social Sciences. And today we're in the lecture by the course co-organizer and that will kick off the final leg of the course, which is Avel Guinin-Carlu's lecture, Social Constraints and the Existential Stance, What Makes Us Human. So we'll be looking forward to a lecture followed by a discussion. So if you have any questions, please feel free to add them in the live chat or submit to the course syllabus. And also at the end, we'll talk about how we're going to conclude this course. So with that, thank you to you, Avel. Hello. So do you see the slides correctly? Everything is fine? I will take silence as a yes. So the role of this session is basically to show you the inspiring intuition behind the whole course and to show very concretely, well, very abstractly, but in my mind concretely, why the specific formalism of Active Inference and Social Constraints is really going to understand the human species and the structural evolution of, you know, things that we do. So we'll talk of social constraints of the existential stance and of what makes us human. So disclaimer, this paper, this presentation is closely aligned with the upcoming paper, which was name is like the same, I think. It's tentative for now. And whose quarter, whose job is a co-author. So first, let us ask basic questions about the topic at hand. What is culture? By psychologist, Seshila Haiz, culture is typically used in one of three ways. Culture can be information or behavior required socially. So like, if I am an ant and I figure out that you need to go somewhere because of the pheromones, this is culture. It can be socially acquired behavior that is shared within a group. So for example, if I am a nape and I figure out how to open knots with a stone thing and the group, just everyone does it the same way. This is culture. And it can be the adaptive evolution of socially acquired behavior, which is very well. We have evidence of it in the wild for corvids and humans only. So the question is, we know that culture is the human specific, like it is the case simply. But depending on what we mean by culture, it can describe behavior that is very much shared with other species. So the question, a question that needs to be answered is how specific are the human cultural mechanisms? So one core thing to understand is that there is no such thing as the cultural mechanism. The mechanisms of social learning in humans are culturally transmitted, whether you will listen to, I don't know, an elder, a rapper, a scientist depends on the cultural norms in which you exist. And the specific way in which teacher-student interaction are supposed to go is again culturally embedded. And the type of information that people will try to figure out is or transmit is itself culturally embedded. So we can't just write the mechanism for human social learning. This is empirically there is none. And conceptually, I don't think the notion is coherent to begin with. So one core element for this is we have basically explicit linguistic rules about who we who we should trust, like listen to your parents. And those are linguistic and language varies from culture to culture. So this changes from culture to culture, pretty straightforward. And in general, the practice of cultural transmission of intentional teaching, they are varying between cultures. And I do not want to get

into it because I don't think I would say something that is accurate without notes. And we can, however, take a look at

a behavior that is, I think, nearly universal, which is storytelling. When people tell a story, they use stories to transmit norms. And like, in the specific culture of the Agda, or Agta, I have misspelled it at least once, I think it is Agda.

Like you have a core role of storytelling in telling people what to do. And you have specific tales that were studied and that were

associated to specific norms. And these are very important because like, here, we have laws that are written somewhere, and that, like people in blue, will come and hit you if you don't follow them. But this is not something

that is universal. So in a context where you do not have states, you typically have no really other way to transmit norms than, you know, talk about it. And so from there, everything is culturally embedded. Like the way you

are talking the norms, the kind of norms can be transmitted given this way. The way specific cues will be understood given a specific

niche, et cetera, et cetera. So yes, all this to say, there is no big C culture. There is cultural transmission, there is cultural

evolution, there is cultural evolution, but there is no specific mechanism or, you know, natural type of culture in humans.

But there are categories of behavior that are very obviously human specifics, and that are very obviously cultural, as in direct

contribution to culture. For example, fire, it's obviously majorly important to culture, because if you have fire, you get a lot of calories.

You get a lot of calories, you get a big brain, like you cannot fall in the brain. But fire, you know, it doesn't produce culture.

That's not how anything works. Things that do work directly to produce culture are language and role-taking.

So language, I think you are familiar with it. You are using it now on the receiving side.

It is a practice that basically embeds communicative intents into a specific set of symbols that can be letters, that can be words, like oral, it would be phoneme, I guess. I don't know if phoneme or the word is a symbol, but you have symbols and they embed meaning and people use it. And what is critical is that each of the symbol

has a public meaning, has a public like semantics and pragmatics. If I tell you chair, maybe we don't know precisely what specific thing enters in the chair category, maybe your perceptual.

The session of chair will vary, but we agree on the basics of the chair, we agree on what we can do with chairs, we agree with a lot of things implicitly by the use of this word. And we can use the rule of syntax to manipulate this unit of meaning in a specific way. I sit on a chair, I'm talking about an action that is related to a specific object that is a chair. And you understand what I mean. If I was trying to, you know, mimic it, maybe I would get it through. Maybe you would think I'm saying I'm crouching over a bed. Why not? And so the language and the fact that it does a publicly instantiated meaning is critical in, you know, keeping a coordination in the specific thing that people

do and why they do it, which is unambiguously cultural. And you have another more subtle mechanism process, I don't know, phenomenon, which is shared intentionality. So by the work of mainly Tomasello, we know that chimps are not very social and they can learn, they are, but as intelligent as humans, but at least young humans, but they usually are not really interested in what other chimps are doing. And most importantly, they are not interested in, like, sharing goals, trying to figure out what is the end results that some other chimps is upholding and trying to like mesh in this plan to help him. While in human from a very own edge, we are very motivated and we are very motivated to like figure out what we, whatever we are collectively doing and figure out what one's role is in that and do that. And this is likely not, you know, a bona fide cognitive function that stands on its own, but it's a behavior that is a cognitive process that is that is anemic in humans, very rare if existing everywhere else. And that is core to Morse theory of the transmission of culture. So these two things we know are human specifics up to maybe very small, important details, important details, and we know are critically embedded in what we call culture. And so what I will do from now on is to provide a specific theory of culture. So the active inference of cultural landscapes that, in my opinion, basically explains or at least meshes well with why those two specific phenomena are so important from the appalling of culture. But it also means that culture is not information per se. It's something that is more organic and that evolves more like, you know, biological systems themselves evolve than population of genes. And from this account, we will go to figure out what it means about collective organization, figure out why we have gods and taxes. And from there, we will go to a more physical and physiological consideration and ask how basically this theory can be written within physics through the active inference formalism. So let us go. First, I'm just presenting the theory of the active inference of clear landscape and I'm presenting a core concept, the concept of constraints that derive from at least my treatment of that. So active inference, probably a lot of you know what this is, but this goes on YouTube, so I will re-explain. Active inference is a theory of cognition where basically agents enact expectation of the world. They have a world model and they make predictions over the world model and they strive for those predictions to remain coherent with what they actually observe. And that means that the agent will, you know, update beliefs that are false. If I hear me-oh and I think there is no cats I will then think that it is more likely. And on the other hand, I will try to provoke, to realize a prediction that I cannot need. So for example, if I really want to hear me-oh, if I really expect to hear me-oh and I expect this to be like a basic staple of my life, I will maybe figure out how to get a cat. But that works not well with cats, but with drinking or eating, I expect to be a thing that has water in it because else I die. So when I'm thirsty, I end thirst by drinking. And we have this basically duality of predictive, like normal, what we call typically perception, just trying to figure out what there is outside and updating our belief when we don't. So I said perception, but it is more inference. And there is active inference, which is

like the realization of this inference through action. And the core argument of active inference is that those two are the same process. They are the same process that is entailed by an information geometry. So basically a space of belief over which the dynamics of an agent unfold. And the core feature of this approach is the dissolution of agency, as we typically understand it.

We typically understand agency to be like, hey, I'm an agent. I want to do this. And I will reason about what I want. And I will reason about how to get it. And I will do the thing that I reasoned about. Reason is important. It's not the basic cognitive function that analyze everything we do.

I do not think the statement is controversial. And in active inference, the model we have is that like at multiple scale, in multiple contexts, we get prediction over our sensor rotor flow, and then strive to realize those predictions. So what that means is that our actions are not constrained by will. In the general case, they are constrained by what looks more like a self model, a model of oneself, and what oneself would typically do in contexts such as the one I'm exposed to.

Related concepts is the notion of skill and intentionality. So this is this skill intentionality thing is a theory that stands at the interface between active inference and ecological psychology, which I will absolutely not get into. And the core argument of skill intentionality is that when you are a skilled agent, so agents that understand how to do things in the world, you do not experience the world as a represented space in 3D with 3D models in it. And each object is well defined and has a lot of properties that are embedded in a world model that you love in a video game. What you see is a landscape of basic opportunities for action. I will see a computer, I know I can mash the keys. I think you say keys. And I know I can go to the internet and I know I can do this or that. I see a

door. I know I can go through it if it's open. I can close it. I can go through it if it's closed and I can open it.

This is the basic opportunities that the door presents. And I don't say a logical consequence, but a very coherent

entitlement of active inference is that I will see minimal model of things around me that are based on the opportunities for action that they present. I will orient my attention minimally to grasp what I can do with things and what the fields, the perceptive fields that I live in is a landscape of those affordances.

So the picture is an extract from Fallout New Vegas, which is the greatest RPG. Please play it. This is important. Only thing you need to keep in mind on this course. And that shows you like a radio.

In like 3DF, you know, radio. And you have an action button that tells you whether you want to activate the radio. And you have HPs and APs. I have no idea what AP is. And this looks more like a naive field than what you get into a really realistic presentation of like a picture, photo picture of your environment. Because you have a perception of the object. Like you have a 2D array that presents, you know, your vision of an object. But it's embedded in things that tell you where you're looking at, things that tell you what you can do with things, things that tell you what are your states, what are your basic states when you're engaging with this object. And this is, I think, pretty nice illustration. And this is a very interesting illustration of what we could mean by exception and the landscape of first, what is that you're being transmitted. You need less.

You should.

Sorry, Apple, could you repeat the last 30 seconds there? It was just like it hung up.

Okay, just repeat the last 30 seconds. Thank you.

So this is a picture from Fallout New Vegas, which is a good game. And I claim that it is a pretty good illustration of what we talk when we talk about action-related perception. Because you see the object and with slash through the object, you also see what opportunities for action in presence. In this case, activate the radio. And you see it in a way it is embedded with your physiological states. So the HP and AP. And a core feature of that is that when I experience affordances, I feel directly through the perception whether the affordance is good or bad. If I see, I don't know, a cliff and I can fall off the cliff and die, this will cause fear. The perception of that possibility of falling and dying will be negatively loaded. If I see, I don't know, a hot bed in winter and I know I can now rest after a day of work when I was cold, I will be happy about it. I will pursue the possibility in a like positively balanced way. And so this is the case for like basic physical physiological thing.

But this is also the case for cultural norms and values. We come to experience the world as a world of affordances. And this is the same for the social world. I will feel bad or good depending on what possibility for action I will experience in a given context. And um,

this is a picture of a forest path near a ecological site that is, that is studied by the research group that funds me, Xscape. And like from a pure direct perception kind of way, there is like a very complex statistical picture that are made in the picture. But if you are a human, a thing that walks, what you will perceive like feel directly, experience directly is just a possibility to walk along the path and follow it. And this is something that you feel is a property of the path. I think you see the path, you think I can go through a path. You don't think there is a 3d space. And in this 3d space, there is a set of statistical feature, one of which being a flat thing in the middle. And among the entire space of action that they have, one of them is to walk through the path. Like just do this and this and this and this and this. You just feel like you can go through there.

And the way humans learn about each other, the way humans commit and come to expect things about each other, they can develop basically a sense of what is culturally possible or not possible in a given context. And this sensation, this perception of cultural possibilities, it is embedded in the way we pay attention to things. We will pay attention to different things according to what is culturally relevant in a given context, in a given situation. And this is a core argument of the thinking through the minds account of cultural evolution, which was spearheaded by Maxwell Romstedt and Vesier, I forgot his first name. And because of everything that we saw before, because of the dissolved agency that we have underactive in France, the cultural patterns in the flow of attention, they will not simply be experienced as abstract norms and value that one could or could not respect. They will effectively constrain the thing you can pay attention to, the outcome you expect in a way that will drive reaction towards specific outcomes. And so this cultural landscape of affordances, they embed intentionality and normativity. As in the patterns that are relevant to explain normativity and intentionality, they are in the landscape, they are not like only at least in your brain cells.

[illegible]

All right. Could you repeat the last 45 seconds?

Yeah.

So what I'm saying is that the engagement with cultural landscape parentheses, it is due to the existence of shell constraints which act over individual behavior.

And instead of looking at how each individual agent engage with the landscape, I can look at the constraints that emerge from this collective activity.

Okay.

Yeah.

Yeah.

And so let us get back to the question of what makes us human.

Those were language and role-taking.

Language provides a space of public representation, like representation in the classical cognitive sense, that you can negotiate that are actually out there in the literal sense.

They are sensory signals that are in the environment.

And that's structural and Asian behavior.

So when you talk to yourself, it will constrain what you pay attention to implicitly, what you can and cannot do.

And when you talk with others, this will influence what you say and what you want to say, and what you perceive to be relevant to say.

And so it will constrain your behavior.

And in a much stronger and more robust way than you could have to implicit intentionality inference.

And role-taking in a more subtle way brings agents to think existentially,

to not ask the question what I am supposed to do in this situation,

but what an agent that is embedded in the specific cultural context where I am embedded is expected to do.

And so this basically forces you to have a synchronization between what you experience as possibilities for action

and what you expect are cultural expectations of our behavior.

So these two mechanisms would be very, very strong mechanisms to enforce the coherence of the cognitive landscape between agents.

And it just so happens that there are things that exist only in the one species that has full-fledged, cumulative, open-ended cultural evolution.

So yes, I think this is a...

interesting.

Okay.

Avel.

You just unshared your screen.

Sorry?

You unshared your screen or is that done?

No, you did not unshare my screen.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

Yeah, thank you.

a duality between each agent engaging with the landscape for dance and constraints that exist over the landscape that is shared that is straightforward but in the set of constraints that is characteristic of a given landscape basically if to account for planetary culture evolution this must be a set of constraints that changes to time that builds new constraints and new chemical flow and it's something that you cannot get with a landscape in the sense of chemical system theory that is not the ecosystem theory does not deal with systems that do open-ended evolution so the question is what relation can we make between the dynamics of enculturation according to which agent integrates specific cultural expectation and the dynamics of an ended culture evolution through which cultural landscape and the constraints like constructs itself through time and this is where we get to the more holistic side where i think we can see why the specific construct of constraints is a critical as compared to the simple cultural landscape approach let's take a look at this there is a thing called called biological autonomy it is a quite recent term in its formalization the relevant formalization that describes the ability of a living system for self-construction like maintaining one structure and challenging it through developmental and evolutionary time and the best formalization we of that so far is from monteville and mosio and it's a constraint closure a constraint is simply a relatively slow physical process that shapes a faster process like canalize material or information or energy flow without itself being affected during at the time scale of the interaction and for monteville and mosio you get biological autonomy where you get a set of

constraints that collectively produce itself each constraint is produced by other constraints of set and each constraint produces other constraints in the set this is a very simple id this is a very informal id in the sense that it is not doesn't have dynamics doesn't have a physics it is just a formal statement of what we mean by autonomy

and this id basically is the best grounding we have to think about what is living and what is an agent although in itself it does not entails agency and when we look at social systems or early systems as a set of constraints

what you can see is that systems such as the brain multi-scale organized system they can be read as basically a set of nested constraints like your dml the activity of your default mode network which is the basal activity of the brain in the absence of stimulation

it's uh basically is considered of chemical constraints over what the composants um tissues do at a given point in time

and this itself is a constraint over what individual cells do uh through time which itself is a constraint over the evolution law of an individual cell etc etc etc etc

so i can always look at a multiple scale system as a set of nested constraints at nested scales

and in systems that are um autonomous that are biological cognitive agents um not only the um smaller scale uh systems like uh cell

they can uh they enable they construct and they enact the constraints that are irscale agents but irscale agents can also shape back

shape back the behavior of low scale agents like of course if i have you know if i do something which is something that i as an agent do this will constrain my activity of my neural networks this is like heavy and learning this is common to all models learning learning goes through change in network structure and so there is an opportunity to model uh emerging agent agent from like collective network activity and to look to look at the interaction between the description at nested scales of behavior

and this is applicable very straightforwardly to the human species so this is uh

uh surrounding states and something that you can look is that uh it is very dense it has very dense buildings that are built in a specific way it is near a river uh which enables the um trade at great distance it has a bridge so entry to the city is uh constrained to a specific point and it is essentially uh surrounded by uh surrounded by uh several fields very flat lands that are rubble that are um fertilized probably by uh water from the river and basically all of these features are enabled the production of first state societies societies with a central administration that does stuff and these societies will robustly produce context uh material context that favorize their own survival

for example dance settlements to enable you to like have everyone together to tell them what to do temples which give a much grounding to the authority of the states uh bridges that allow you to move things around and to control what gets in and out and more cultural several fields that allow you to like take the food and bring it back to the city and the production of such settlements and such material environment by states is uh somewhat robust intercultural feminine

at a more like ancient or multicultural scale um you have a set of processes that are critical in the construction and action of norms in humans and most likely uh common ancestor to the

those processes is dancing and singing dancing and singing that they give you um like embodied synchronization and this is shown to drive development of group identity and this gives you basically uh emotional grounding of the norms that are shared within society because they those norms they are not abstract expectation they are embedded in your specific emotions in your specific embodiments while you do things that are uh critical to your ecology and ritual activity and storytelling they allow uh more complex self-normativity by using you know symbolism and the system of normativity they are maybe less embedded embodied sorry so less embedded in the uh like one's uh direct experience but they allow a more more complexity in the nature of the norms that are uh transmitted and enforced

and uh the these norms the by the simple fact that they are perceptible through material symbolism and through storytelling they become a causal force in the cultural landscape in the that is shared by a group member so they become thing that are uh the power to produce or constrain the framing in which people will read

specific situation and therefore constrain and produce social behavior and this um this is like i've said this already so this looks like repetition but i want to hammer this because if this is the case the case uh then we can be interested in what complex uh tails in sorry in agency that emerges from the constraints embedded in complex tales so we know that uh people can perceive non-material agents under certain

conditions uh this is common in uh schizophrenia like you will hear voices uh that tells you to do things and that are not embedded in a specific body it is common in a decisive valentine disorder because you will have agents that basically share your body while while having an existing while not being tied to a specific uh body like one sorry i'm uh misspicking one specific personality can exist while they are not in control of your body and so in a way that is not contingent on the direct material and action of what they want and want not to do and under a lot of ventrology anthropologically um described conditions uh that are not considered

pathological you can basically have regimes of attention that produce enormous religion that you can talk with and that have motivation and that have immediate traits uh which are often called uh talpas which is a term borrowed from buddhist literature and uh so this emergence of the material agents how to say if i consider an agent to be something a closed set of constraints and i consider things like tales

and ritual as mechanism that produce constraints in behavior there is basically no difficulty whatsoever at conceptual level there is no unexpectedness or no complexity to say that uh specific systems are constrained that happen to be closed and that are embedded in those tales and rituals they exist as biological agents and uh so we kind of basically a model of gods of non-material agents which is a better less like the way to put it that relies on a systematic training of your regime of attention and of course uh when it's collective it will like those agents will be social agents they will be things that have uh power that have the ability to frame the debate around specific things if you and i think that uh so this is uh sorry the painting that is on the right of the screen is a painting by william blake

that presents uh basically a late 18th i think notion of god which is uh not really old testament which is a bit more new testament but which is a muslim sensation of the um a clockmaker god that was popular at the time and what you see uh on the painting is iris and so william blake's version of god that is in the sky like very powerful very jose like uh imagery and that is using uh like this his hands to draw a compass that is considered a symbol of um like um constructing things of masonry like the compass is one of the freemason uh symbol for the same reason it is a symbol of uh orderly creation in william blake's uh paintings and so if we share a belief that there is a god somewhere and he can he build the world as it is intentionally and this god is benevolent or at least much too powerful for us to fight with basically your option uh they boil down to following the order because if you don't god will remove you from the order and you don't want that do you so even if you don't have the existence of like um material material material sensation of this god even if people don't really really believe that there is actually a guy with a bird that does stuff uh they shared uh ritualized uh belief in an entity that has this specific motivation and these specific traits it will uh shape heavily the social landscape that is available to everyone and this is compatible with um although it does not entail the existence of actual agency that is tied to this specific entity through the constraints that this entity concretely is and something that i find again interesting is that religion that are popular today so christianity islam buddhism hinduism if you think of something as a religion it will likely fall under this this um handle are typically what white house called doctrinal religion which are religion which are religion that are where basically you sit down and you look at rituals and there is a priesthood that is trained to do the ritual in a specific way that is supposed to mean specific things and this is basically optimal to build very robust notions of uh what gods are what they want from you what they do what they're motivated by etc etc

so i must insist that this is with a question mark but in principle the um the strong constraints model it can produce a touristic theory of gods and natural agents sorry as socially embedded non-material agents that are computed uh collectively through engagement with a specific cultural landscape where uh ritual uh ritual and doctrinal um um elaboration are are uh important components and i can say with uh exclamation mark that this uh model it enables naturalistic theory of social agency that is not tied specifically to individual bodies if i consider a social system uh actual organization to be autonomous that is closed system of constraints uh zain agency or maybe um non-trivial agency it simply reduces to the power to shape sorry shape this landscape and you can have uh emergent social things do that you can have states which follow specific rules so social constraints and specific norms so against constraints that have autonomy and that are able to shape the landscape but in all of this like this part in my opinion uh shows that social constraints specifically are relevant to study these social worlds that they correspond to relevant social processes in human order agents that are institutions and maybe again question mark uh non-material agents uh sorry uh intelligence into institution so gods uh the question that needs to be asked is how we can model the portion of constraints because all of these rely on the idea that somehow in the landscape that we experience there is such a thing as constraints and they are well-defined

thing that exist over collective behavior and that they are the pro the property to construct itself and for reason we'll very shortly see to frame this mathematically in a coherent way is insanely hard i don't know how to do it no one knows how to do it we have leads but they are leads nothing else and so we'll turn to the question of uh whether active inference frames a gauge theory

foreign consideration you should know what this is by now god's theory we will see that soon so to explain gorge theory i need to explain explanation which is fun but mostly symmetries uh in physics in physics uh we frame physical problem around symmetries uh symmetry it is uh algebraic so something that has composition like plus groups of transformation over the space of possibilities which are irrelevant from the perspective of the same dynamics it is very abstract so let's look at symmetries this is the ceiling of the lotfola mosquay nisfarne what you can see is a symmetry under the C_{32} symmetry group if you shift this picture by um one over 32 uh full turn it doesn't change this is what we mean by a symmetry this is a transformation a set of transformation that is closed and that does not change well not a picture of course in the context of physics but the dynamics of the relevant system we are studying and for formal reasons i will not absolutely not get into here explanation in physics it usually derives from the statement of symmetry for example when i state that energy is conserved this is the same formally as stating that my system is symmetric in time when i state that momentum is conserved that radius formally uh to stating that my system is asymmetric in space okay so most things in physics can be framed as simply symmetries over system dynamics

and a gauge theory is simply uh elaborate instance of this core id so if your symmetry is uh has two properties first it is continuous so not like the uh a set of small rotation but like the set of all possible rotation that you can take uh around the circle and local so you can apply the symmetry transformation to every single pointy space uh your problem basically can be framed with a random degree of freedom so because the symmetry does not change your system uh whether your system where your stem is in the for example clockwork rotation it is not important it does not change the dynamics so this redundant degree of freedom we call a gauge parameter

um the core feature of gauge theory is to take the geometry of that problem and make it into a physical force so um when i say uh gauge parameter so this is um you should see in uh in the screen uh non-trivial z over two z possible bundle of a circle which means basically that you have a circle it has two stages and when you are over it you can be either on top or down and you can't know where it is so this is uh not quite the graph freedom because you have only two choices to get to the graph freedom you have to get a continuous space of choices but this is basically what we mean by gauge parameter it is a geometric property of your system that you know mathematically exists but that cannot possibly be observed or how to say corresponds to a choice in how you observe your system so this is why we talk of gauge because this is a choice in how we inspire the system but i will not get into the details of this here and the thing is that because of its redundancy the gauge parameter it does not have to be the same in a specific trajectory uh at any given point in space and from that you get curvature and curvature if you are anywhere near physics and you know the relativity theory you will know that curve into the like naive embedding space in so if you know

uh vaguely activity theory you know yes sorry just if you're anywhere in physics and then continue from there yes so if you are vaguely familiar with relativity theory you know that uh curvature in a space it corresponds to uh forces so in relativity theory the curvature of space time it corresponds to the gravity force and generally in uh physics the curvature in other weirder gauge spaces it translates in the other fundamental forces so weak force strong force and um electromagnetism so yes just stating that there is a space which has a gauge parameter in it and this space has a curvature it entails that you have a force that emerges over the let us say natural framing of the problem

the problem and uh this uh speaks very closely to active inference so active inference can be framed as a consequence of a formal result that is called the frangy principle let us uh write a chemical system stochastic which means that it has noise in it and which can be a partition between an agent a and environment b and a boundary sorry an environment e and a boundary b such that the distribution of it of the agent and the other elements are independent from each other uh given the discussion of the boundary b which kind of means that any information uh in e that is relevant in a and vice versa has to be in b but doesn't quite mean that but it's very close then you have information geometry which is uh like geometrical construct where each point in the agent corresponds to a distribution of belief over uh the space of external states and because of this information geometry each possible internal state of the agent it entails implicitly a belief over external states so this means that given the condition of partition ability i can write bayesian belief over my environment as a simple geometrical property of the space we are working in and as i just said when you have a geometry and it's curved you get forces and so uh the fep can straightforwardly be framed as a gauge theory where the symmetries in sensory entropy technically but let us say the space of belief i am evolving in directly constrain my action in a specific manner to constrain bayesian coherence between what i expect to be the case and what i experienced to be the case and so straightforwardly this grounds active inference because like what i just said is what i said to explain active inference up to some technical words that were added in but yes this result entails uh predictive processing and predictive control of behavior of the agent and to some extent improves the possibility of this existing in their uh admical system theory that's a very important thing to be able to understand that the fvp can possibly represent enculturation so when i talk of a dynamic called system this is a system with dynamics in it you are i think you understand that i do not think you necessarily understand the uh formalism that we imply by it that we call the system is a flow function ϕ a time parameter t and a state space x so what that means is that to frame an equal system i have to write a space let us i don't know just the euclidean space with three dimension i have to uh specify time which is like naturally in the euclidean space what we think of as time is that we think of as time and i have to specify a specific way in which the um my system will move in space in a given uh step of time so ϕ gives from any position and any time slot where the system will be if it starts from this position after the specific time duration this like this is what a dynamical system is but if you have noticed this this entails that i define preemptively all other space of possibilities of my

system

if i have a system that is a you know open-ended system of constraints that can build new stuff that can be language that can build accounting whatever new states this doesn't go maybe you say uh we could reconstruct a minimal

notion of new stuff happening by work sorry i'll ask him again avil again repeat the last 30 seconds

but your audio is still not there it's like a 30 to 40 second period where your audio just doesn't

transmit none of us can hear it but the internet is fine working over the dynamics of the system percy

learning the ecosystem theory so you can do that either so fine yeah continue yep continue okay

so uh the vehicle system can be written as a flow function f over a state space x and a time parameter t

when you specify x you specify the total space of possibility of the system so if you're a system

that has a penalty evolution in it well you cannot work with a fixed state space you could maybe uh like

circumvent the issue by implicitly having new things happen from a change in the dynamics of the system

but in the vehicle system theory the dynamics are pre-given through the flow function f

the vehicle system system system system is a specific deterministic flow so you can't change the flow

through time so you can't model the construction of new constraints and you can't model the emergence of

new parameters that affect causally your system so the f in of itself cannot present the creative

evolution of either cognitive kinds such as such as money shares or constraints

the possibility of the system that you can't change the flow function of the system

and this is where we have to get in um weirder uh spaces formally to get something that looks like

you could model open-end evolution although i cannot say definitely it can because we didn't yet

by construction when we work with a set theory which is the basic theory of mathematics which is about

specific objects that are predefined we can't have what uh cough band calls adjacent possible which

corresponds to new evolutionary possibilities that are brought about by the evolution and evolution of

evolutionary system like living systems um simply set theory forces you to define all of the possibilities

at step one of the modeling process so can't have your system that builds new possibilities with activity

this is not a coherent concept in set theory however what we can do is to look at rather than set theory

at a category theory which is let us say another approach to mathematics which looks not at specific

objects and their relationships through functions but of specific spaces and their relationships to

functors and this change in approach basically does not entail at least not a priori that when you write

a system you have to write all of the possibilities that are within the system preemptively you can write

a space and then look at the relationship between the space and other adjacent spaces that could correspond to adjacent

possibles and one thing that is related to that at least in spirits is uh i won't get into detail but formal approaches that allow you to build quantum state spaces

from the property of quantum observation rather than the opposite so simply speaking we can look at the problem of

how cognition brings about new possible not as a space of possibilities in which there is a flow

but as a set of spaces in which specific events such as observation such as enculturation such as

invention of new concept can bring about new spaces and when we rewrite active inference in the language of

category theory we get things that are that do not quite say the same thing as chemical system theory actual uh actual uh actual frangy principle uh but we get things that are very very complicated so the diagram that you see on the bottom left of the screen is a concom diagram i could tell you what the concom diagram is but i don't know so i could not explain that is meant to present the alignments in reference frame or the mutual active inference between two subsystems so again formally we have no reason to believe this can't model of

an evolution but also given the level of effectiveness of the math we have very very few grip with which to understand what we're looking at and how this relates to specific cognitive and cultural processes so there is a lot of progress to be made in that direction before we have a self-standing theory of open-ended cultural evolution and the basic handle of some constraints i think is a good bridge between uh the very concrete very simple chemical system theory and the very abstract um problem of many evolution so let us wrap this up here and uh try to take out a few snippets of actionable information that was included in the talk uh first agents by nature they engage with the world with their world as a cognitive landscape that is defined by the modalities by which they perceive the world that is defined by the intrinsic normativity that the agent experience and that is

again sorry just go back 22 seconds

again

we'll breathe before we jump into the conclusion

okay

i still don't hear you

we'll see

We can do the slide game where someone else presents the slide.

Any volunteers?

Okay.

Yeah, I heard that.

Okay.

Cool.

Yep.

Please continue.

Okay.

We'll try to go fast because this is not agreeable.

So one, agents engage with the world as a cognitive landscape, which is defined by our forenses.

So minimal representation of the action that are possible in the landscape and the normativity that's embedded in the possibility of the action.

Two, the fact that humans collectively enact a relatively coherent structural landscape, it entails that we have social constraints over collective behavior.

And the existence of those social constraints, they enable, although do not necessitate, the emergence of higher scale agencies such as institution, organization, and maybe godlike beings.

And last, the dynamics of constraints, they entail a duality between the creation of meaning.

So the way we humans create a way to understand the world and the construction of physical possibilities, possibles in the sense, specific concrete causally relevant possibilities that characterize a specific social culture organizations.

And maybe like physical social practices, but just don't worry about it.

And all of it is embedded in this diagram, the engagement and the duality with a system of constraints that produce itself in a panning manner.

So a claim that I'm making on the record here is that if we understand correctly every arrow in this diagram, we understand culture evolution, we understand what human are, and we understand panning evolution in the cultural world.

And here I'm making on the right.

And here I will go through the slides in case someone needs a reference and has the time to, you know, go through a video to find the reference.

So thank you and sorry about the audio thing.

Did this go well?

Yes, it gave some much needed silences, so it was all good.

Cool.

Well, first, Mal or Maria, if you would like to give any comments or reflections.

Otherwise, we can go to a few topics and then talk about how we want to conclude with the course.

But first, Mal or Maria, do you want to say or ask anything?

No, I mean, this was a great summary.

I think you covered a lot of grounds.

I think ideally I would have liked to see how much we can do with gauge theory and how this can potentially link us to.

Understandings of power.

I think that would be that would be interesting.

Cool.

Thank you, Maria.

All right.

See you, Maria.

so that we'll then know exactly who wants to present what,
and then we can ask how many minutes they want to do or something like that.

But Mau?

Yeah, I think it would be good to get people who are interested in getting more active
to come and make a small presentation, explain what they understood,
describe the highlights for them, and potentially link it to some topic in social sciences
we may not have covered here, but that they now see through a different lens.

I think that would be interesting.

Okay.

All right.

I will look through the live chat while I was getting some water,
and then we'll ask any questions that people come up with.

Is there ever symmetric information between or among social entities?

Or maybe from the scripts perspective, Mau, or from the, or however you see it in relation with the
constraints?

I mean, what about asymmetric constraints?

I'm not sure I understand the question.

Can the person give a little more detail in terms of the question?

Okay.

I will look back.

But there was a question in the chat about, can information ever be symmetric?

You talked about symmetries.

So are those symmetries in the information that two social entities have?

Or are they in the constraints that entities experience?

But what are the symmetries and what are the breaks?

I would say that depends on which space you model the problem.

Information entails symmetry because it has to be readable.

So you have to be a conservation between context or time of something
that is considered to be the content of the information-bearing thing.

But also it is asymmetry-breaking because if you take any presentation format, let's say binary,
a random string means nothing.

And you have to break the symmetry of the uniform distribution to say something.

So it depends very heavily on how specifically you are modeling the problem.

And then you have to be a mathematical representation.

But for fundamental reasons, any mathematical representation depends on the implicit model of the
symmetries of the system.

Because you want to present things that do not matter, but you have to present things that do matter.

And then you have a very uncomfortable amount of freedom over how you want to present any single thing.

Certainly this comes up in learning active inference and in talking about it, like what you described, how many ways there are to model a given thing,

which is related to which symmetries are preserved as you mentioned.

So how do we reckon with that space and with the different ways of representing the same social setting?

So a reason why I'm not quite comfortable with the notion of information is that in all possible settings, where I define quantum information theory as not possible because like very few people think about it, information will be a probability distribution.

It will be like a functional of our probability distribution that is minus to the logarithm of the probability.

That is what self-information is.

And this is defined over probability distribution, where distribution is defined over state space.

So implicitly information entails the differential of state space.

So if you want to work over things that are prior to state spaces, you cannot work with information like any coherent definition.

So basically the one thing you could do that is vaguely inspired from work in quantum information theory, mainly by Christopher first, is to not look at the probability, but at the, sorry, not to look at the information.

I'm using that term in a colloquial way now because I have to switch meaning.

Information not as a property of the distribution, but the property of quantum operator that corresponds to observation.

And that embeds basically, we could say again, in a very, very colloquial sense, something akin to an expectation or a measurement or a synchronization event in the active inference model.

And this for a specific type of operator, for a specific type of space, you can get basically the space that flows from the operator rather than the other way around.

And then information is like, it's not a Kolmogorov complexity.

It's not shallow information. It's not a variant of entropy.

It is basically what is hypothesized by the specific operators that you use to interact with matter and with cultural and material environments.

And maybe, maybe there is a big maybe here.

I should say maybe one more time.

Maybe this can happen in some way to account for the process of incarceration where the, basically, the process of encapsulation,

is considered as the integration of specific reference frames, expectation colloquially.

And this entail the evolution of the cognitive landscape and our social constraints.

Either defined over the cognitive landscape or defined as a self-standing mathematical object.

I do not know.

But if we have that, and if we have a decent level of understanding of what constrains the evolution of neural fence frames,

we are the self-standing theory of the unfolding of cognitive landscapes and of open-ended cultural evolution.

This is a sufficient condition, I am sure of that.

But it is very, very hard to reach.

Are you seeing the window?

Yeah, I'll bring it in. That's fine. Thank you.

Okay. No, no.

So, the dissolution of agency is basically the idea that when we think about agents, people, like we think as a system that has a body and that has a brain and that has the ability for discussion, like deliberating over what you do and not do.

And we place agency at the level of discursive beliefs.

So, if I say I want to grab the water and I grab the water, I grab the water out of my agency.

If I inadvertently, like I don't know, bump into something and make it fall or grab without thinking about it.

Is it attentive?

Not. Most people would say yes, but that's actually not entailed by the most used model of agency.

Like, legally speaking, you are held as not responsible or less responsible of things that you did not do.

have the opportunity to liberate over.

But from a purely cognitive perspective, and if that inference model is correct, the bulk of the intentionality of your ability to direct your own flow that you have does not come from reasoning and deliberation and central executive system, by cognitive function that we associate with reasoning and agency.

It comes from the basic expectation you have yourself and the world, because your expectation over the world's constraint heavily implicates the opportunities for action that you perceive,

and your beliefs over yourself constraint just as heavily the options that you will end up pursuing.

And so the agency, your ability to decide off stuff, it mostly reduces to proactive process over your self-model and your world model.

And this is, while not like, I don't see any problem formally or empirically with that picture.

Like, as I understand it, that comes very well for the, for the neuropsychiatric and like psychological literator and sociological literator.

Like this is good, this works scientifically.

But this is counterintuitive because, again, most people will expect you to have reasoning and then to decide off stuff and then to execute all the stuff you decide.

Which, again, I don't think this happens.

So, how our question is the information geometry.

So, the information geometry is a geometry that is defined within a system equipped with a partition, blah, blah, blah.

And so, by a geometry, I mean basically a space of curvature that exists over your bezel space.

So, this is basically the same thing that I'm presenting in the bottom left of the screen here.

You have a space that is your normal space where you see stuff happen, which is represented here by the line.

And you have a curvature that is defined by some of the things that are defined over the space.

And by the differential principle, when you have this partition condition that is verified, then each space, sorry, each state in the agent's internal states, like each specific point it can be, corresponds to a belief distribution

over the states of the environment.

And so, this is in this sense that we have information geometry.

Like we have a geometry, so a curvature over space, where the space is the internal state of the agent and the geometry is belief over external states.

And furthermore, this matters because the agent is actually constrained by the image.

The agent is effectively constrained by the geometry in such a way that its internal flow optimizes the Bayesian coherence of the belief that exists according to the geometry.

So, this is highly math. This is like straightforward, but lots of words, lots of perception, no real visual way to put it.

So, this is more something to like read about and something to hear about, which is why gloss over it.

But I hope going over it again brings a bit more clarity.

How would you connect what you just said to the differences with written and oral culture?

Like what is hearable? What is seeable? How does writing in the printed or etched or digitized form relate to how you discussed culture today?

I see you as frozen, Daniel. Can you hear me?

I can hear you. Mal?

Okay. So do answer or do Mal answer?

Mal, if you want to. Yeah, please.

I can give a little bit of an idea, but effectively we're talking about precision and temporality, right?

So writing and discourse.

The discourse has to relate to pragmatics and the discourse is likely to be interpreted because it has lower precision than say writing has over time.

So here you're talking about the information geometry will be bent and shifted through the precision of the embedded normativity that is given rise to through these markers that you can point to over and over and over with high precision.

I think that's the link.

Yeah, basically that's. So when I talk about role taking and language, I claim that the core reason why this matter is because this enforce coherence between the cognitive landscape experienced by agents and writing changes massively the scale at which it is possible.

And it changes massively the level of emotional embedment of the language.

So it is much lower in speech than sorry, much higher in speech than it is in writing, especially if you account for like singing as a force of speech.

So historically writing very clearly and very ambiguously powered the rise of administration.

So states, so organization that do a lot of accounting and bookkeeping, like book is literally in the word, and that's enabled the centralization of resource management.

So this is explaining in the logic of writing by Goody, G O O D Y.

And yeah, basically this writing is a core cultural innovation that enables centralized policies, city states, things like that.

And that's the goal of writing.

And that's the goal of writing.

And that's the goal of writing.

And that's the goal of writing.

Yeah, that's very interesting.

So one topic, Avil, that we've discussed, and maybe you could unpack a little bit here, is thinking of culture and these cognitive ecosystems leads to consideration of the cognitive security or of different information risk aspects.

So where does that come into play with the formalism or what you discussed here?

So from a cognitive security aspect, one thing you can read in the formalism quite directly is that if you control the environment, as in the cognitive landscape experienced by agents, you control what the agent do directly.

So there are two things that are very, very, very stereotypically associated with cognitive security.

This is Russian propaganda, not quite propaganda, cognitive warfare campaigns, pretty much around the world.

More generally, the propaganda that is used by authoritarian regimes to get stuff going.

And the cults that historically are tied to the worry of brain control by evil cults that can control you from afar, like you are the telecommands.

And so in both cases, actually, let us get back a bit when you are a state that does propaganda.

And like when you are a state, you can do a lot of material engineering.

So producing the environmental states of the in which people live in.

And for example, if you look at the city of Brasilia, which is one of the very few specifically built capital city that exists in the world, there is like one in there is Brasilia and there is like, I think maybe four others.

And those cities are basically very big glorified administrative centers and Brasilia in particular, it has neighborhoods for work, neighborhoods for leisure, neighborhoods for for habitation.

It is extremely divided.

And because of this dividedness, you basically don't get to do stuff that are not work or hang out with your family.

That is not a possibility.

And of course, this changes massively the way people experience their life.

And this is massively the topology of this network that can emerge.

But when you are doing a cognitive warfare, like in the sense that NATO defines, so you are a state or state like organization, and you are trying to fuck with and

other states or state like organization, you don't get to just build cities that are bad.

But it's not a possibility that you have.

So what the Russians do in a systematic way is to try to basically create doubts over what information is reliable, so that people feel paralyzed.

The target, I must insist critically, is not the information that is available.

It is trust that people can have the action ability that the people can give to the information.

And so they tend to propagate a lot of conspiracy theory nonsense because it creates a lot of space and it prevents the possibility for public debates to occur.

And another strategy they use is to manipulate the personal identity that are embedded often in ethnicity or in the case of the US, of advanced countries that have

where political life is mostly decoupled for ethnicity.

And I understand the irony to say that this is the case in the US, but you don't have like ethnic, ethnically specific parties or armed forces.

So this is less the case than elsewhere.

So in these countries, they will use political identity.

So they will use Republican identity or leftist identity, but mostly Republican identity to just push complete nonsense and get people to act in a way that overall destabilize the group of the US over their own population.

And so this is the basic way that we do collective warfare.

And you can understand this in terms of social constraints, because again, the target is not like information in itself.

It's the possibility that is afforded by the social context in which we evolve at an instant.

The goal is to create doubts and noise in the public setting, straight constraints that are not compatible with efficient collective organization.

And a way like a way where you do it is when you see it being done in a much more systematic and efficient way is, of course, with interest state influence, where state administrators and state institutions, they try to create grip over a population.

And if you look at the birth of nationalism, like we have the notion today that nations are somewhat coherent political groups that, let us say, the French hang out together and take decision together.

This notion, it's not quite recent, strictly speaking, but the fact that it is dominant, that it is just a standard expectation is very recent.

And it's it derives from the French that somehow where a group of administrators somehow managed to convince everyone that they shared identity and they shared mutual obligation responsibility in virtue of this identity.

And the way in which this happens is they sent institutors in villages and they sent soldiers and you want to listen.

The children had to listen to the instructors, explain to them that they were French and they needed to do French things like going to war for France.

And if you would not let the children like do that, you went to prison and no one wants to go to prison.

So with, of course, I am I'm doing a very, very huge caricature here.

This like there is not a shadowy set of influence that sense.

This is a over-specification, but what happened is that the French state managed to create a social niche up in a very, very foreign villages where French hood became a basic condition of existence.

Most of your social thing, the fact that you were legally recognized as existing, the activities that you had, a lot of those were embedded in your perception of being French.

And so when this is the case, and when you see that administration tends to be on top of things, the administration tells you you have to go to war in virtue of your Frenchness.

Then you go to war, you don't like fight against the administration, you fight against whoever the mission tells you.

The fact that you have agreed implicitly that you are French with friends makes this a lot harder to resist

basically than if like the state just sends soldiers at you and kidnap you and give you a swear and tell you to fight.

This is not how you make things efficient.

Armiras this way are unorganized, they are unmotivated, they are not good.

And if you want people to have stakes, you need to give them power.

Or you need to give them the impression they have power because they are French and here French govern things.

And whether you are in a set of French that actually direct things or not, bad question to ask.

Just go to the trenches.

And this type of things we would not think as community warfare because we would not think as a big thing.

As a between.

Like implicitly when we talk about warfare, we talk about between states or between nation conflict.

And this is within a nation within a group.

But you can look at cults and how they create social influence.

You can look at, for example, one of the most famous cults that existed, which is the Moonies or the Unification Church.

They were widely accused to make brainwashing.

They were widely accused to somehow control people and make them do things they don't want to do.

That is not a possibility you have as an organization.

Like you cannot kidnap people and hope that it will be okay.

You will get killed by police if you do that.

Most likely in the US, in this specific space where the Unification Church evolves.

What you can do is to create a context where you have a huge grip over what people can and cannot say, what they can and cannot think about themselves.

You can tell them that your leader is the coming of Jesus that is going to bring everyone to paradise.

And you can make people go to work sessions to talk about it.

And you can just like physically cling to those people all the time until you are quite confident that they feel that whatever you are doing is important.

And you can have very big emotional ritual where everyone like cries and become extremely embedded in the group membership.

And you can be extremely supportive of group members while being extremely like aggressive to anything that go away from group identity.

And when you process this environment and people and go to this environment for a long time, they tend to develop this group identity.

So when you tell them the big leader tells you to give us 2000 euros, sorry, 20,000 euros for your ancestors not to go to hell.

That seems reasonable.

That seems like a reasonable ask, which it would not unless you had a very tight control over the social environment of someone.

Because then it's not just giving money in exchange for a good.

This is about upholding yourself as an entity as someone who follows and believes the cult's teaching.

And yet again, this is something that is formalized directly in terms of through constraints.

Because this is about what you can and cannot do in a given certain context.

This is not about like abstract beliefs.

It's not about brain control.

It's not about anything but what you can and cannot do in a given social context.

All right. There was a ton there.

Let me let me kind of retrace a few of the pieces that I heard there.

So towards framing cognitive security under this cognitive constraint framework that you're discussing, for which active inference under the free energy principle is the process theory that gets it done.

First, you described how modern states have a lot of administrative capacity and affordances, especially through writing and especially again in the last decades and centuries of information environment changes.

And then you pointed to two kinds of patterns or genres or motifs of influence or cognitive warfare, propaganda and cults discussed one example of them surely could have gone into more detail or there's other examples and perspectives and so on.

And as Mal alluded to earlier, precision modulation is a key aspect of constraint.

I mean, precision modulation of how tightly you're holding the baseball bat is really the movement.

It's precision in wobble is going to be related to the to the grasp.

And so increasing uncertainty can lead to anxiety and doubt.

All of that's very wrapped up in a narrative setting.

Then you discussed how there's a ratcheting of thinking and acting like a state, breaking symmetries like states do and getting grasp over the subject over the citizen.

That's a complex topic, but that's what you went.

And then there are asymmetries around control of the environment, the information environment, ability to to read, write, edit in the information environment.

And that leads to these engineered distributions of what is important or what means what, which justifies or even writes the blank check for requests of funds, whatever the narrative is around those funds.

But more fundamentally, those strategies can threaten sense making, which is to say the inbound perceptual component of active inference.

And then also decision making and action selection, which is the fundamental complement, the outbound aspect of active inference.

I'll read another comment from the live.

Yes.

Oh yeah.

Do you want to add anything else on that?

Yeah.

Like I'd like to point out a very, very specific example that is contemporary, which is the campaign by Israel to remove people from the Gaza bans.

So just, I don't know when people will watch it.

So on the 7th of October, 11 days ago, Hamas, which is a terrorist organization, Islamic organization that is dominant in the Gaza bans attacked Israeli and slaughtered Israeli settlers that were nearby and slaughtered tourists that were nearby and attacked military bases.

They made, I think 2,000 deaths.

And so the state of Israel as a counter that is currently led by very far right parties, things that we would not have equivalent in of in Europe or the US has decided to answer by blaming each individual Palestinian for this and trying to deport them elsewhere.

And so like this, this is not itself attempting to comment about what I'm trying to comment about is that every single comment you will see about it on Twitter makes the confusion between Hamas and Palestinian on one side and IDF slash Netanyahu and the Israeli on the other side.

And so this is a framing that says, basically, people are trying to decide whether the Hamas action are justified or not.

And implicitly in this debate, they create a context where whoever lose, whichever group lose the legitimacy debate gets to be killed to the last.

to the last.

So that produces a framing that makes the only possibility in the conflict explanation because each side, like there is no, like this is not a two side dynamic.

There is no Palestinian and there is the Israel and there is the, I will not make all the relation.

But by this framing, like the framing that either the Arabs or the Jews must prevail and kill the other, you give a lot of power for groups to just make this a genocidal war and justify things like cutting water to one million people because like 200 of them committed a massacre.

And like, I do not know if there is a single actor that tried to get this.

I do not think this is the case, but this is a very powerful instance of how propaganda does not need to target what you believe, but it does need to target how you frame problems.

Yeah.

A lot to say on that.

Any last pieces you want to discuss?

No, it is good.

I have talked a lot.

I'd like to stop talking.

Okay.

Well, thank you.

We will send out updates to people about the coming weeks.

So thank you.

All right.

Bye.

Ciao.

Thanks for your attention.

Goodbye.

Bye.

Bye.